Construction Chemicals

SUBMITTAL DOCUMENT

O-Patch C

Fibre-Reinforced Polymer-Modified Repair Mortar



Overhead	Strength	Bond	Thickness
No sag	≥ 45 MPa	≥ 2.0 MPa	5 – 40 mm

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

O-Patch C — Fibre-Reinforced Polymer-Modified Repair Mortar

Project Name		**Document No.**	OMX-MS-OPATCHC
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Patch C	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Patch C on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers preparation, mixing, application, curing and inspection of O-Patch C (fibre-reinforced polymer-modified repair mortar).

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after adding water.
SSD	Saturated surface dry — substrate damp but without standing water.
Layer thickness	Thickness of one application pass.
Curing	Keeping the applied material protected from rapid drying until it gains strength.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.

Role	Responsibility
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Patch C	25 kg bags.

5.1 Key product data

Property	Typical Value
Density	2.0 – 2.1 kg/L
Water demand	13 – 15 %
Pot life	30 – 40 min
Setting time	3 – 5 hours
Compressive strength, 7 days	≥ 30 MPa
Compressive strength, 28 days	≥ 45 MPa
Flexural strength	≥ 7 MPa
Bond strength	≥ 2.0 MPa

6. Tools & Equipment

- Forced-action or slow-speed mixer
- Buckets
- Trowels
- Floats
- Formwork
- Measuring jugs
- Curing materials
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Goggles
- Dust mask
- Safety shoes

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Protect bags from moisture; do not use material with lumps.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Break out damaged concrete to sound substrate.
2. Clean exposed rebars to bright metal and protect as specified.
3. Pre-wet to SSD.
4. Pre-wet absorbent surfaces to a saturated surface-dry condition; remove standing water.
5. Protect adjacent surfaces with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Check the batch number and expiry date and record them.
2. Pour 13 – 15 % water into the bucket.
3. Add powder gradually and mix 3 minutes.
4. Mix only the quantity that can be applied within the pot life.

11. Application

1. Press the mortar firmly onto the SSD substrate.
2. Build up in layers of 5 – 40 mm.
3. Finish with a float.
4. Work in manageable areas and finish each area in one operation where possible.
5. Check layer thickness and consumption regularly and record the results.

12. Curing & Protection

- Cure for at least 3 days; protect from sun and wind.
- Protect from sun, wind and rain; keep moist where required during the first 24 hours.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before application	Prepared, sound and clean as specified	QA/QC
Climate	During work	Application temperature: +5 °C to +40 °C	Site Engineer
Consumption	During application	1.8 – 2.0 kg/m² per mm thickness	QA/QC
Finish	After curing	Uniform, no defects	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Shrinkage cracks	Rapid drying; layer too thick; excess water	Remove, reapply in thinner layers and protect from drying
Poor adhesion / hollow areas	Dusty or dry substrate	Remove, clean and pre-wet; reapply
Dusting surface	Excess water or late finishing	Light sanding and apply a fresh thin layer

Defect	Likely Cause	Remedy
Uneven finish	Insufficient trowelling	Sand and apply a finishing layer

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear gloves, safety goggles and a dust mask when mixing.
- Contains cement; alkaline when wet. Avoid skin and eye contact.
- Avoid inhaling dust.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: wash with soap and water; cement is alkaline when wet.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools and mixers with water immediately after use.

17. Limitations

- Application temperature: +5 °C to +40 °C.
- Substrate SSD.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate prepared and approved	☐ Yes ☐ No	
Batch numbers recorded	☐ Yes ☐ No	
Material mixed as specified	☐ Yes ☐ No	
Applied at specified consumption	☐ Yes ☐ No	
Curing and protection carried out	☐ Yes ☐ No	
Final inspection passed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

O-Patch C — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Repair and Grouting Materials



O-Patch C

Fibre-Reinforced Polymer-Modified Repair Mortar

Product Description

O-Patch C is a fibre-reinforced, polymer-modified cementitious repair mortar for vertical and overhead repairs of beams, columns, slabs and soffits. It is applied in 5 – 40 mm layers and usually needs no bonding agent.

Uses

- Beams and columns.
- Slabs and soffits.
- Patching and levelling.
- Spalled concrete.

Advantages

- Vertical and overhead application.
- 5 – 40 mm per layer.
- 28-day strength ≥ 45 MPa.
- Bond strength ≥ 2.0 MPa.
- Fibre-reinforced.

Application

1. **Surface Preparation**
   Remove damaged concrete; clean and treat rebars; pre-wet to SSD.
2. **Mixing**
   Add 13 – 15 % water and mix 3 minutes to a uniform paste.
3. **Application**
   Apply by trowel or hand in 5 – 40 mm layers; cure.

Consumption

1.8 – 2.0 kg/m²
per mm thickness

Packaging

25 kg bags.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Density	2.0 – 2.1 kg/L	—
Water demand	13 – 15 %	—
Pot life	30 – 40 min	—
Setting time	3 – 5 hours	—
Compressive strength, 7 days	≥ 30 MPa	—
Compressive strength, 28 days	≥ 45 MPa	—
Flexural strength	≥ 7 MPa	—
Bond strength	≥ 2.0 MPa	—
Layer thickness	5 – 40 mm	—

Consumption

Thickness	Consumption
10 mm	18 – 20 kg/m²
20 mm	36 – 40 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Application temperature: +5 °C to +40 °C.
- Substrate SSD.

Application Notes

- Protect exposed rebars with O-Zinc I where specified.
- Apply in layers of 5 – 40 mm.
- Scratch each layer before the next.
- Cure for at least 3 days.

Health & Safety

- Wear gloves, safety goggles and a dust mask when mixing.
- Contains cement; alkaline when wet. Avoid skin and eye contact.
- Avoid inhaling dust.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: wash with soap and water; cement is alkaline when wet.
- Inhalation of dust: move to fresh air.
- Ingestion: rinse mouth and seek medical advice.
- Clean tools and mixers with water immediately after use.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.